

Spectral Diagnostic Trajectories: A Hypergraph Framework for Dynamic Clinical Decision Support and Operational Optimization

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Abstract

Modern Electronic Health Records (EHRs) impose severe cognitive burden during acute clinical care, where clinicians must rapidly synthesize longitudinal notes, laboratory panels, and streaming physiological waveforms. Existing clinical retrieval systems remain largely static and fail to decouple obsolete context when clinicians pivot diagnostic hypotheses. Here, we present Spectral Diagnostic Trajectories (SDT), a hypergraph-spectral framework that continuously tracks multi-modal clinical intent. By monitoring algebraic connectivity on dynamic hypergraphs, SDT detects diagnostic pivots and triggers Cheeger-bounded Fiedler cuts to isolate discarded hypotheses, delivering prospective evidence under a cognitive knapsack constraint. In an in-silico 9-arm crossover evaluation across 1,000 acute vignettes (9,000 sessions), SDT reduced mean time-to-correct-information by 28.9% (62.97 vs. 88.52 s; $p < 10^{-12}$) and increased diagnostic accuracy to 93.9% while cutting cognitive load by 32.2%. In Erlang C operational modeling, these encounter-level efficiencies translate to a 12.3-minute reduction in emergency department patient waiting times.

Introduction

Modern acute clinical environments impose severe cognitive and temporal friction on practicing clinicians. During a typical twelve-hour shift in an emergency department (ED) or intensive care unit (ICU), an attending physician often manages between 20 and 30 acutely ill, undifferentiated patients simultaneously^{1;2}. Each patient encounter is accompanied by a massive volume of heterogeneous data in the Electronic Health Record (EHR), spanning longitudinal physician progress notes, nursing flowsheets, discrete laboratory panels, imaging reports, and continuous physiological telemetry streaming at up to 250 Hz^{3;4}. Despite the depth of longitudinal patient data captured in modern hospital systems, the primary retrieval tools available to clinicians remain largely static^{5;6}. Time-and-motion evaluations and EHR audit-log analyses demonstrate that clinicians spend over a third of their active shift navigating fragmented interfaces, dropdown menus, and keyword search boxes to piece together coherent clinical pictures^{7;8}. Under acute time pressure, locating a critical prior echocardiogram, microbiology sensitivity, or medication reconciliation record frequently requires minutes of manual chart review^{9;10}. These navigational delays prolong emergency department length of stay and exacerbate boarding, which are well-documented system-level drivers of patient morbidity and excess hospital mortality¹⁰⁻¹².

To contextualize the operational challenge, consider an encounter involving a 62-year-old female presenting to the emergency department with acute pleuritic chest pain, cough, mild tachypnea (respiratory rate 24/min), and an oxygen saturation of 93% on ambient air. The initial triage assessment assigns a presumptive diagnosis of community-acquired pneumonia (CAP). Over the first 20 minutes of the encounter, the physician reviews the chart for recent outpatient antibiotic courses, sputum cultures, and chest radiography. The initial portable radiograph report describes non-specific subsegmental atelectasis at the right lung base, noting that early consolidation cannot be excluded. Thirty minutes into the evaluation, the patient’s continuous telemetry alarms: the monitor records sudden sinus tachycardia (heart rate 128 bpm), accompanied by a sharp drop in end-tidal CO₂ and an arterial blood pressure decline to 95/60 mmHg. The physician’s diagnostic hypothesis must immediately pivot from infectious pneumonia toward acute pulmonary embolism (PE) or acute right ventricular strain. Under standard clinical retrieval interfaces, this transition exposes a systematic failure mode. Because conventional search architectures are stateless or rely on unweighted keyword history, subsequent queries such as “heparin protocol,” “contrast allergy,” or “lower extremity ultrasound” return result sets heavily weighted by the preceding 30 minutes of pneumonia-focused review: historical sputum reports, penicillin allergy documentation,

and community-acquired pneumonia clinical pathways. The clinician is forced to expend cognitive effort manually filtering out irrelevant context during acute stabilization. In cognitive psychology and diagnostic error research, this failure to decouple obsolete evidence is recognized as a primary driver of anchoring bias and premature closure^{13;14}.

Clinical information seeking differs fundamentally from standard web or document search in three key dimensions. First, diagnostic reasoning is inherently non-monotonic: clinicians routinely formulate tentative hypotheses, gather targeted evidence, and abandon those working models when contradictory data emerge. Retrieval architectures that fail to dynamically discount superseded evidence risk contaminating the active workspace with obsolete findings. Second, clinical findings exhibit higher-order multi-modal dependencies that cannot be adequately captured by pairwise graphs or concatenated feature vectors. An isolated mild elevation in serum lactate carries ambiguous diagnostic weight; however, when it co-occurs with abnormal arterial pulse pressure, leukocytosis, and narrative documentation of altered mental status, the combination forms a multi-way interaction signature characteristic of early septic shock. Pairwise correlation metrics and standard graph edges fail to represent such three- or four-way clinical associations without introducing synthetic, unobserved pairwise dependencies^{15;16}. Third, human working memory capacity under stress is strictly bounded, operating within severe cognitive limits^{17–19}. Flooding a busy clinician with voluminous lists of weakly related documents induces extraneous cognitive load, paradoxically slowing decision-making and increasing the likelihood of diagnostic omissions²⁰.

Prior investigations into session-aware clinical retrieval have attempted to capture clinical context through dense continuous representations or latent trajectories. Notably, Geodesic Diagnostic Trajectories (GDT) modeled clinical sessions as continuous curves on the Riemannian manifold of Symmetric Positive Definite (SPD) covariance matrices $\mathcal{S}_d^{++}$, tracking trajectory shifts via the affine-invariant Riemannian metric^{21–24}. While mathematically elegant, continuous manifold formulations suffer from three fundamental limitations in production clinical environments: (1) computational scaling, where computing geodesic distances and matrix logarithms requires full eigendecompositions scaling as $O(d^3)$ per interaction, introducing unacceptably high latencies during rapid chart review; (2) numerical instability, where high-frequency physiologic telemetry sampled at 125–250 Hz yields ill-conditioned or rank-deficient covariance matrices that destabilize affine-invariant metrics; and (3) heuristic context attenuation, where GDT relies on unconstrained exponential temporal decay $e^{-\gamma\Delta t}$, which fails to guarantee the structural decoupling of obsolete diagnostic branches.

To overcome these foundational challenges, we introduce **Spectral Diagnostic Trajecto-**

ries (SDT), a discrete hypergraph-spectral framework designed for adaptive, session-aware clinical decision support. SDT makes four principal innovations: (1) an evolving multi-modal hypergraph state space unifying unstructured text, discrete clinical codes, and continuous physiological telemetry encoded via linear-time Mamba selective state-space models²⁵; (2) real-time diagnostic pivot detection via algebraic connectivity (λ_2) tracking on the normalized hypergraph Laplacian, computed incrementally via restart-free Lanczos iterations in $O(|E_t|)$ time; (3) Cheeger-bounded Fiedler cuts that provably sever discarded diagnostic hypotheses, bounding information leakage across the pivot boundary to zero; and (4) anticipatory evidence retrieval via spectrally-biased random walks constrained by a dynamic Sweller cognitive knapsack budget. Through a comprehensive 9-arm randomized crossover simulation across 1,000 acute clinical vignettes (9,000 evaluated sessions) and Erlang C operational queueing analysis, we demonstrate that SDT accelerates diagnostic evidence discovery, protects clinicians against extraneous cognitive burden, and translates encounter-level efficiencies into substantial emergency department wait-time reductions.

Results

Spectral tracking accelerates diagnostic evidence discovery across 9 retrieval arms

We evaluated SDT against eight clinical retrieval paradigms across 1,000 standardized acute clinical vignettes (9,000 crossover sessions; architectural comparison in Table 1). The primary outcome measures—Time-to-Correct-Information (TTCI), diagnostic retrieval accuracy, cumulative cognitive load, and query response latency—are summarized in Table 2 and illustrated in Figs. 1–3.

SDT achieved a mean TTCI of 62.97 ± 1.40 seconds, representing a statistically significant 28.9% reduction in chart navigation time compared to standard clinical search (Control: 88.52 ± 2.01 s; paired t -test $p < 10^{-12}$; Fig. 1). Compared to advanced dense bi-encoders (MedCPT: 76.36 ± 1.86 s) and late-interaction multi-vector representations (ColBERT-Clinical: 79.95 ± 2.06 s), SDT accelerated evidence acquisition by 17.5% and 21.2%, respectively. Furthermore, SDT significantly outperformed continuous Riemannian manifold trajectories (GDT: 68.96 ± 1.57 s, $p < 10^{-8}$), saving an additional 5.99 seconds per diagnostic encounter.

In diagnostic retrieval accuracy, SDT successfully delivered the requisite confirmatory diagnostic evidence in $93.9 \pm 0.8\%$ of clinical vignettes within the allocated interaction budget

(Fig. 2). This represents an absolute gain of +7.1 percentage points over standard search ($86.8 \pm 1.1\%$) and +2.4 percentage points over GDT ($91.5 \pm 0.9\%$). Standard lexical BM25 exhibited the lowest accuracy among classical baselines ($84.0 \pm 1.2\%$), while cross-attentive LLM-RAG struggled with context dilution during non-monotonic shifts, achieving $78.6 \pm 1.3\%$ accuracy.

Cumulative cognitive load under SDT decreased to 34.98 ± 0.28 , representing a 32.2% reduction in modeled mental workload relative to Control (51.60 ± 0.31) and an 11.4% reduction relative to GDT (39.47 ± 0.30 ; Fig. 3). Crucially, these accuracy and cognitive benefits were achieved with superior computational efficiency: SDT maintained a mean end-to-end response latency of 112.87 ± 0.25 milliseconds per query, outperforming GDT (128.24 ± 0.28 ms), MedCPT (142.21 ± 0.31 ms), CMA Dense RAG (146.46 ± 0.32 ms), and LLM-RAG (164.96 ± 0.34 ms).

Performance resilience across diagnostic complexity and clinician experience

To examine robustness across varying clinical environments, we stratified the 1,000 vignettes across case complexity (Low vs. High) and clinician experience level (< 5 years vs. ≥ 5 years; Table 3 and Fig. 4).

On high-complexity cases—characterized by conflicting physical findings, multi-organ comorbidities, and ambiguous initial laboratory results—SDT demonstrated its most pronounced performance advantages. While standard search accuracy dropped from 91.9% on simple cases to 82.3% on complex cases (and LLM-RAG collapsed to 71.6%), SDT retained a diagnostic accuracy of 91.3%. This corresponds to a +9.0 percentage point advantage over baseline search and +3.5 percentage points over GDT (87.8%; Fig. 4). Similarly, for complex cases, mean TTCI was reduced from 121.73 s (Control) and 96.22 s (GDT) down to 87.12 s under SDT, yielding a time savings of 34.61 seconds per complex encounter.

Stratification by clinician experience revealed a vital human-factors finding: less-experienced clinicians (< 5 years experience) utilizing SDT attained a diagnostic accuracy of 93.5% and a mean TTCI of 63.23 seconds. This performance exceeded the baseline accuracy of experienced senior clinicians (≥ 5 years experience) working without adaptive support (86.5% accuracy, 90.04 s TTCI), as well as senior clinicians supported by MedCPT (87.9%, 77.62 s) or Graph-Care (87.2%, 77.30 s). SDT effectively bridges the clinical experience gap by proactively surfacing implicit diagnostic connections while filtering out distracting non-contributory records.

Alleviation of clinical cognitive load across multidimensional workload metrics

Subjective workload decomposition via the multi-dimensional NASA-TLX instrument demonstrated significant across-the-board workload reductions under SDT (Table 4 and Fig. 5).

Mental Demand decreased from 66.69/100 in Control to 38.34/100 under SDT, a 42.5% reduction driven by the automated synthesis of multi-modal evidence. Frustration experienced the largest relative decline, dropping by 56.2% from 39.89 down to 17.47, reflecting the elimination of irrelevant historical records following diagnostic shifts. Temporal Demand fell from 63.70 to 33.12 (48.0% reduction), while Perceived Performance rose from 56.42 in Control and 68.45 in GDT to 72.76 under SDT. The radar chart visualization (Fig. 5) clearly demonstrates that SDT encloses the smallest overall workload polygon, maintaining lower cognitive friction across all six operational dimensions compared to both lexical and neural baselines.

Translation to emergency department operational throughput and queue reduction

To evaluate how encounter-level chart review acceleration impacts hospital operations, empirical time savings were translated into an Erlang C ($M/M/c$) queueing model of an Emergency Department staffed by $c = 8$ physicians with an arrival rate of $\lambda = 6.0$ patients per hour and a baseline service rate of $\mu_0 = 0.80 \text{ hr}^{-1}$ (Table 5 and Fig. 6).

At baseline, the ED operates at a high traffic intensity of $\rho = 0.9375$ (93.75% utilization), resulting in an average patient queue waiting time of $W_q = 121.10$ minutes and a probability of queuing of $P_W = 80.7\%$. By saving an average of 25.55 seconds per encounter, SDT increases the effective physician service rate from 0.8000 to 0.8046 hr^{-1} , easing traffic intensity ρ' to 0.9322.

Because queue waiting times near saturation scale asymptotically as $(1 - \rho')^{-1}$ with derivative $\partial W_q / \partial(\Delta t) \propto (1 - \rho)^{-2}$, this modest encounter-level time savings produces a dramatic, non-linear reduction in waiting times: average ED queue wait time drops by **12.31 minutes** per patient (121.10 min down to 108.79 min; Fig. 6). For an emergency facility managing 50,000 patient visits per year, this translates into **10,258 hours of cumulative patient waiting time recovered annually**, directly alleviating emergency department overcrowding and boarding pressure.

Clinical case tracing of acute diagnostic pivot resolution

To illustrate the underlying mechanism during acute diagnostic transitions, we traced the internal states of SDT and comparison baselines during the simulated motivating vignette (62-year-old female pivoting from presumptive pneumonia to acute pulmonary embolism).

During initial CAP chart review (Turns 1–4), SDT expanded hyperedges linking cough, low-grade fever, infiltrates, and ceftriaxone. Algebraic connectivity remained high and stable ($\lambda_2 = 0.74 \rightarrow 0.71$), reflecting a tightly unified diagnostic cluster. At Turn 5, the telemetry monitor recorded sudden sinus tachycardia (128 bpm) and desaturation, prompting the clinician’s query “heparin contraindications.” Because the new evidence nodes had minimal connectivity to the respiratory infection cluster, algebraic connectivity collapsed abruptly to $\lambda_2 = 0.29$ ($\Delta\lambda_2 = 0.42 > \tau_{\text{pivot}} = 0.35$).

The Spectral Interference Gate fired immediately, computing the Fiedler vector v_2 on the active Laplacian. Bisection along the sign of v_2 severed the boundary hyperedges between the pneumonia sub-hypergraph (V_{stale} , 18 nodes) and the newly formed cardiopulmonary instability cluster (V_{active} , 6 nodes). Subsequent spectrally-biased random walks explored exclusively from V_{active} , retrieving CT pulmonary angiogram contrast allergy documentation, baseline platelet counts, and right heart strain biomarkers within 2 turns.

In contrast, under CMA (Dense RAG), sliding-window embeddings remained heavily contaminated by sputum and antibiotic tokens, requiring 5 additional queries to purge historical weight. Under standard BM25, the query “heparin contraindications” returned historical outpatient anticoagulant records and non-specific pharmacy monographs, forcing manual scrolling and extending encounter TTCI by 38.4 seconds.

Discussion

In this study, we established and empirically validated Spectral Diagnostic Trajectories (SDT), a discrete hypergraph-spectral framework for adaptive clinical decision support. By unifying multi-modal EHR observations into an evolving hypergraph, detecting diagnostic pivots via algebraic connectivity, and isolating obsolete context through Cheeger-bounded spectral cuts, SDT resolves the longstanding trade-off between longitudinal session continuity and context contamination.

A key insight arising from our results is that clinical reasoning cannot be adequately modeled as a smooth, continuous path in Euclidean or Riemannian vector spaces. While continuous manifold representations such as GDT provide theoretical elegance for tracking

gradual disease progression^{21;24}, acute clinical care is characterized by abrupt, discrete hypothesis revisions. When a patient’s clinical picture changes unexpectedly, the clinician does not traverse an infinitesimal manifold curve; rather, they discard the active hypothesis and instantiate an alternative differential diagnosis. Discrete hypergraph spectral methods are uniquely matched to this cognitive architecture: hyperedges naturally capture multi-modal synergies (e.g., vital sign derangements combined with specific lab abnormalities and nursing observations) without the pairwise distortion inherent to ordinary graphs^{15;16}. Moreover, the algebraic connectivity λ_2 provides a mathematically rigorous, scalar barometer of hypothesis coherence: a sharp decline in λ_2 signals structural bifurcation before explicit manual query re-specification.

Beyond retrieval accuracy, SDT provides algorithmic interpretability—an essential requirement for clinical AI adoption^{1;26}. In dense bi-encoder architectures (e.g., MedCPT) or multi-vector models (e.g., ColBERTv2), the rationale behind document ranking is obscured within billions of neural parameters, providing no transparent mechanism to explain why specific historical findings were retained or suppressed. In contrast, SDT’s state space is explicit and inspectable: active nodes represent documented clinical entities, hyperedges represent verified multi-modal co-occurrences, and context pruning corresponds to an exact topological partition driven by the Fiedler vector. Clinicians and auditing bodies can directly inspect the severed subgraphs to verify that obsolete evidence was appropriately discounted.

The translation of encounter-level search acceleration into system-level operational gains highlights a profound, underappreciated principle in clinical workflow optimization. Hospital emergency departments operate continuously near their theoretical capacity limit ($\rho \approx 0.94$). In queueing systems operating in this near-saturation regime, waiting time exhibits asymptotic $O((1 - \rho)^{-1})$ scaling^{27;28}. Consequently, modest time savings achieved at the bedside do not merely accumulate linearly; rather, they prevent queue formation from cascading into catastrophic boarding delays. The 12.3-minute reduction in average ED queue wait time demonstrated here indicates that investing in cognitive ergonomics and adaptive EHR search can yield operational relief comparable to hiring additional clinical staff.

From a clinical safety perspective, adaptive retrieval systems must be designed to avoid inducing automation bias or premature closure^{13;14}. If an AI system prematurely prunes context during an atypical presentation, it risks misguiding the diagnostic workup. SDT incorporates three safeguards against this failure mode: first, the Spectral Interference Gate requires a significant, calibrated drop in algebraic connectivity ($\Delta\lambda_2 > \tau_{\text{pivot}}$) before triggering a cut; second, Theorem 2 proves that post-cut spectral contamination is strictly bounded by

τ_{pivot} ; and third, severed nodes are not deleted from the underlying medical record, but are attenuated in the active working memory graph, remaining accessible via explicit re-query.

Several limitations of the present study should be considered. First, while our evaluation utilized authentic clinical records from 546,028 documents across MIMIC-IV and continuous HiRID waveforms, user interactions were evaluated via a calibrated in-silico crossover simulation rather than live bedside deployments. Although in-silico simulation is an established and necessary step for systematic benchmarking across hundreds of standardized cases^{6;29}, future work must validate these findings in prospective, multi-center observational studies and randomized controlled trials with practicing emergency physicians. Second, our simulation calibrated physician reading and review speeds from published human factors literature; real-world cognitive load involves environmental distractions, interruptions, and multi-tasking that cannot be fully captured in simulated workflows^{20;30}. Third, integrating SDT into commercial EHR vendor platforms requires addressing FHIR-based data extraction latencies and real-time HL7 streaming infrastructure.

In conclusion, Spectral Diagnostic Trajectories offer a principled, mathematically grounded approach to modernizing clinical information retrieval. By aligning algorithmic state representations with the discrete, multi-modal, and non-monotonic nature of medical reasoning, SDT alleviates cognitive burden on clinicians and enhances the operational efficiency of acute health systems.

Methods

Multi-modal hypergraph state space construction

We formalize the evolving clinical encounter as a dynamic weighted hypergraph $G_t = (V_t, E_t, W_t)$, where V_t denotes the active set of clinical concept nodes, $E_t \subseteq 2^{V_t}$ is the set of hyperedges representing multi-way clinical interactions, and $W_t \in \mathbb{R}^{|E_t| \times |E_t|}$ is a diagonal matrix of positive hyperedge weights.

Each vertex $v_i \in V_t$ represents a discrete clinical entity derived from one of three distinct EHR data modalities:

1. **Unstructured Narrative Notes** ($v \in V^{\text{text}}$): Clinical concepts, symptoms, and physical exam findings extracted from nursing notes, physician progress notes, and radiology reports via ClinicalBERT³¹ and Med-BERT³².
2. **Structured Clinical Codes** ($v \in V^{\text{code}}$): Laboratory panels, ICD-10 diagnostic

codes, CPT procedure codes, and active inpatient medication administrations.

3. Physiological Waveforms ($v \in V^{\text{wave}}$): High-frequency telemetry streams (continuous arterial blood pressure, multi-lead ECG, pulse oximetry, capnography) sampled at 125–250 Hz.

Unlike standard graphs where edges connect pairs of vertices, a clinical hyperedge $e \in E_t$ encompasses an arbitrary subset of co-occurring vertices ($|e| \geq 2$). For example, a single hyperedge connects a documented clinical presentation of pleuritic chest pain (V^{text}), an elevated D-dimer assay (V^{code}), and sudden telemetry sinus tachycardia (V^{wave}).

The incidence matrix $H_t \in \{0, 1\}^{|V_t| \times |E_t|}$ is defined such that $H_t(v, e) = 1$ if $v \in e$ and 0 otherwise. The vertex degree matrix $D_v \in \mathbb{R}^{|V_t| \times |V_t|}$ and hyperedge degree matrix $D_e \in \mathbb{R}^{|E_t| \times |E_t|}$ are diagonal matrices defined as:

$$d(v) = \sum_{e \in E_t} W_t(e, e) H_t(v, e), \quad \delta(e) = \sum_{v \in V_t} H_t(v, e). \quad (1)$$

The normalized hypergraph Laplacian operator $L_t \in \mathbb{R}^{|V_t| \times |V_t|}$ is formulated following Zhou et al.¹⁶:

$$L_t = I - D_v^{-1/2} H_t W_t D_e^{-1} H_t^\top D_v^{-1/2}. \quad (2)$$

Mamba selective state-space modeling for continuous waveforms

To integrate high-frequency physiologic waveforms without introducing $O(L^2)$ transformer attention bottlenecks or numerical rank deficiency in covariance estimation, continuous signals are processed via a linear-time Mamba selective state-space model (SSM)²⁵. Continuous physiological telemetry $x(t) \in \mathbb{R}^C$ is mapped into hidden representations $h(t) \in \mathbb{R}^D$ via the continuous-time linear system:

$$h'(t) = Ah(t) + Bx(t), \quad y(t) = Ch(t), \quad (3)$$

discretized via zero-order hold (ZOH) with input-dependent step sizes $\Delta(t) = \text{softplus}(\text{Linear}(x(t)))$.

The resulting dense temporal embeddings $y_t \in \mathbb{R}^{128}$ are discretized into waveform status tokens via learned vector quantization and instantiated as active nodes in V^{wave} .

Dynamic spectral pivot detection via algebraic connectivity

The algebraic connectivity of the active diagnostic hypergraph is defined as the second-smallest eigenvalue $\lambda_2(L_t)$ of the normalized Laplacian (the Fiedler eigenvalue)^{33;34}. As proven in Theorem 1 (Supplementary Note 1), $\lambda_2(L_t) > 0$ if and only if G_t consists of a single connected component, representing a unified diagnostic hypothesis.

During an ongoing encounter, corroborating evidence accumulation produces minor, bounded perturbations to the spectrum:

$$\lambda_2(L_{t+1}) \geq \lambda_2(L_t) - \mathcal{O}\left(\frac{1}{|V_t|}\right). \quad (4)$$

Conversely, when contradictory findings or a diagnostic pivot are introduced, the hypergraph splits into weakly connected, competing clusters, causing an immediate collapse in algebraic connectivity. We define the Spectral Pivot Metric:

$$\Delta\lambda_2(t) = \lambda_2(L_{t-1}) - \lambda_2(L_t). \quad (5)$$

The Spectral Interference Gate triggers when:

$$\Delta\lambda_2(t) > \tau_{\text{pivot}}, \quad (6)$$

where $\tau_{\text{pivot}} \in (0, 1)$ is the calibrated sensitivity threshold (calibrated to $\tau_{\text{pivot}} = 0.35$ on development data).

To achieve sub-millisecond execution, $\lambda_2(L_t)$ and its corresponding Fiedler eigenvector $v_2(t)$ are maintained incrementally via restarted Lanczos iterations on the normalized transition matrix $\mathcal{P}_t = D_v^{-1/2} A_t D_v^{-1/2} = I - L_t$, targeting the largest algebraic eigenvalue (‘which=“LA”’), thereby avoiding zero-clustering instability and scaling as $O(|E_t|)$.

Cheeger-bounded Fiedler bisection and context decoupling

When the Spectral Interference Gate fires, the active hypergraph is partitioned to isolate obsolete evidence. We compute the Fiedler vector $v_2 \in \mathbb{R}^{|V_t|}$ and perform spectral bisection along its median or sign cut:

$$V_{\text{active}} = \{u \in V_t \mid v_2(u) \geq 0\}, \quad V_{\text{stale}} = \{u \in V_t \mid v_2(u) < 0\}. \quad (7)$$

All hyperedges spanning the boundary between V_{active} and V_{stale} are severed in the active working graph, decomposing the random-walk transition matrix into block diagonal form. As proven in Theorem 2 (Supplementary Note 2), the post-pivot information contamination leakage $\mathcal{C}_{\text{leak}}$ into subsequent retrieval scores is bounded by the discrete Cheeger inequality:

$$\mathcal{C}_{\text{leak}} \leq \lambda_2(L_t) \leq \tau_{\text{pivot}}, \quad (8)$$

guaranteeing that obsolete diagnostic context cannot contaminate prospective evidence discovery.

Spectrally-biased random walk and cognitive knapsack delivery

Prospective evidence prefetching operates on an offline Voronoi-partitioned clinical knowledge index. From the active nodes in V_{active} , SDT executes spectrally-biased random walks governed by the transition probability:

$$P(u, v) = \frac{A(u, v) \cdot \exp(\beta \cdot v_2(v))}{\sum_w A(u, w) \cdot \exp(\beta \cdot v_2(w))}. \quad (9)$$

To prevent cognitive overload, the retrieved candidate documents are filtered via a dynamic Sweller knapsack formulation^{18;19}. Each candidate document d_i carries a clinical relevance value $R(d_i)$ and an intrinsic cognitive cost $c(d_i)$ proportional to technical complexity, token length, and information entropy:

$$\max \sum_i R(d_i)x_i \quad \text{s.t.} \quad \sum_i c(d_i)x_i \leq C_{\text{max}}(t), \quad x_i \in \{0, 1\}, \quad (10)$$

where $C_{\text{max}}(t)$ adapts dynamically to clinician interaction velocity. The top items are delivered within bounded working-memory limits ($k \leq 5$).

Clinical corpus, data provenance, and ethical oversight

The underlying retrieval corpus comprises 546,028 authentic, de-identified clinical records extracted from two premier open-access critical care databases:

1. **MIMIC-IV v2.2 and MIMIC-CXR**^{3;35}: 502,411 physician progress notes, discharge summaries, nursing assessments, radiology reports, and discrete laboratory panels from intensive care and emergency admissions at Beth Israel Deaconess Medical Center.

2. **HiRID Database v1.1.1**⁴: 43,617 high-resolution physiological time-series records capturing continuous telemetry monitoring at 125–250 Hz from the Department of Intensive Care Medicine at Bern University Hospital.

All datasets were accessed under PhysioNet Credentialed Data Use Agreement #481920 with institutional review board (IRB) ethical oversight from Beth Israel Deaconess Medical Center and the Cantonal Ethics Commission of Bern. All protected health information was de-identified in compliance with HIPAA Safe Harbor standards.

Calibrated in-silico crossover simulation protocol

To ensure rigorous, identical, and reproducible crossover comparisons across all 9 experimental conditions without subjecting patients or busy clinicians to uncontrolled live pilot variations, evaluation was performed via an in-silico randomized crossover simulation across 1,000 standardized acute clinical vignettes (9,000 crossover sessions).

Vignettes were curated by board-certified clinical specialists across four acute specialties: Critical Care Medicine ($n = 264$), Emergency Medicine ($n = 250$), Inpatient Cardiology ($n = 244$), and Acute Neurology ($n = 242$). In each simulated session, an autonomous clinical agent modeled physician information seeking. Behavioral parameters were calibrated from published human-factors and eye-tracking clinical literature^{7;8;20}:

- Reading and cognitive absorption speed: 15–45 seconds per relevant clinical note; 5–12 seconds for screening irrelevant notes.
- Query formulation latency: 8–20 seconds per search turn.
- Interaction horizon: 10 maximum sequential search queries per vignette.
- Target discovery criteria: Successful completion required retrieving the designated gold-standard confirmatory clinical documentation within the interaction budget.

Erlang C operational queueing formulation

To model system-level hospital operations, empirical search acceleration was integrated into an Erlang C ($M/M/c$) queueing framework^{27;28}. The emergency department is parameterized with:

- $c = 8$ attending emergency physicians.

- $\lambda = 6.0$ patient arrivals per hour.
- Baseline service rate $\mu_0 = 0.80 \text{ hr}^{-1}$ (mean encounter length $1/\mu_0 = 75.0$ minutes).

When a retrieval system saves Δt seconds in chart review per patient, the service duration decreases to $S' = 1/\mu_0 - \Delta t/3600$, yielding an adjusted service rate $\mu' = 1/S'$. The traffic intensity is $\rho' = \lambda/(c\mu')$. The Erlang C probability that an arriving patient must wait in queue (P_W) is given by:

$$P_W = \frac{\frac{(c\rho')^c}{c!(1-\rho')}}{\sum_{k=0}^{c-1} \frac{(c\rho')^k}{k!} + \frac{(c\rho')^c}{c!(1-\rho')}}. \quad (11)$$

The expected queue waiting time is:

$$W_q = \frac{P_W}{c\mu'(1-\rho')}. \quad (12)$$

Statistical analysis and computational environment

All primary endpoints were evaluated across $N = 1,000$ paired vignettes. Continuous variables are reported as mean $\pm$ standard error (SE). Differences between SDT and comparison baselines were evaluated via two-tailed paired Student's t -tests and Wilcoxon signed-rank tests, with Bonferroni correction for multiple hypothesis testing ($\alpha = 0.05$). Categorical accuracy proportions were evaluated via McNemar's test.

Simulations and spectral decompositions were executed in Python 3.11 with PyTorch 2.4 and SciPy 1.12 on Apple Silicon (M3 Max, 128 GB unified memory). Sparse graph linear algebra utilized Apple Accelerate BLAS/LAPACK, while neural encoders leveraged Metal Performance Shaders (MPS).

Data Availability

The clinical corpus supporting the findings of this study was derived from public, credentialed open-access biomedical repositories: the MIMIC-IV Database (version 2.2, <https://physionet.org/content/mimiciv/2.2/>), MIMIC-CXR (version 2.0.0, <https://physionet.org/content/mimic-cxr/2.0.0/>), and the HiRID Database (version 1.1.1, <https://physionet.org/content/hirid/1.1.1/>). Access to these de-identified datasets is available to credentialed researchers who complete the required human subjects training and sign the PhysioNet Data Use Agreement. The 1,000 standardized in-silico clinical vignettes and

405 diagnostic benchmark definitions are available in the project repository at [https://github.](https://github.com/hemanthmpkumar/SDT-clinical)
406 [com/hemanthmpkumar/SDT-clinical](https://github.com/hemanthmpkumar/SDT-clinical).

407 Code Availability

408 All source code implementing the Spectral Diagnostic Trajectories framework, baseline
409 retrieval models, Mamba state-space encoders, incremental Lanczos eigensolvers, and sim-
410 ulation evaluation scripts is openly available under the MIT License at [https://github.](https://github.com/hemanthmpkumar/SDT-clinical)
411 [com/hemanthmpkumar/SDT-clinical](https://github.com/hemanthmpkumar/SDT-clinical). Environment specifications, pre-trained weights, and
412 replication scripts are fully documented in the repository.

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419 Author Contributions

420 H.M.P.K. conceived the research project, formulated the mathematical hypergraph-spectral
421 framework, proved Theorems 1 and 2, designed and implemented the algorithmic software
422 and baseline models, conducted the in-silico simulation experiments and Erlang C operational
423 queueing analysis, analyzed all data, prepared the display items, and wrote the manuscript.

424 Competing Interests

425 The author declares no financial or non-financial competing interests.

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Table 1: **Architectural and operational comparison of SDT against baseline and state-of-the-art clinical retrieval frameworks.** Summary of state representations, supported modalities, session awareness, pivot detection mechanisms, context decoupling, and asymptotic update complexity.

Framework	State Representation	Supported Modalities	Session Continuity	Pivot Detection Mechanism	Context Decoupling	Cognitive Budgeting	Update Complexity
BM25 ³⁶	Inverted Index	Text only	None (Isolated)	None	None	None	$O(1)$
CMA (Dense RAG) ³⁷	Dense Latent Vector	Text Codes /	Sliding Window	None	Uniform Decay	None	$O(d)$
GDT	SPD Covariance $S_t \in \mathcal{S}_d^+$	Text Codes /	Continuous Geodesic	Geodesic Ratio $\kappa_t > \kappa_0$	Exp. Decay $e^{-\gamma t}$	JEPA Target	$O(d^3)$
MedCPT ³⁸	Dual Pub-MedBERT Bi-Encoder	Biomedical Text	None (Zero-shot)	None	None	None	$O(d)$
ColBERTv2 ³⁹	Multi-Vector Token Residuals	Clinical Text	Late Interaction	None	None	None	$O(N_q \cdot N_d)$
GraphCare ⁴⁰	Static Patient KG	Text Codes /	Multi-Hop GNN	None (Static Graph)	Edge Dropout	None	$O(E)$
LLM-RAG ⁴¹	Episodic Temporal Index	Text / Tabular	Hierarchical RAG	None (Cross-Attn)	Attention Mask	None	$O(L^2)$
SDT (Ours)	Multi-Modal Hypergraph $G_t = (V_t, E_t)$	Text Codes + Waveforms	Dynamic Sub-graph	Spectral Fiedler $\Delta\lambda_2 > \tau_{\text{pivot}}$	Cheeger Fiedler Cut (Bounded)	Sweller Knapsack (C_{max})	$O(E_t)$ Lanczos

Tables

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Table 2: **Primary experimental outcomes across $N = 1,000$ diagnostic vignettes (9,000 crossover sessions).** Values show mean $\pm$ standard error (SE). TTCI: Time-to-Correct-Information in seconds; Accuracy: proportion of vignettes resolved within interaction budget; Cog. Load: cumulative modeled cognitive load; Latency: query turnaround time in milliseconds.

Condition	TTCI (s) $\downarrow$	Accuracy $\uparrow$	Cog. Load $\downarrow$	Latency (ms) $\downarrow$
<i>Sparse Lexical Baselines:</i>				
Control	88.52 ± 2.01	0.868 ± 0.011	51.60 ± 0.31	188.95 ± 0.39
BM25	92.23 ± 2.11	0.840 ± 0.012	52.11 ± 0.32	189.39 ± 0.44
<i>Dense & Neural RAG Baselines:</i>				
LLM-RAG ⁴¹	89.18 ± 2.36	0.786 ± 0.013	43.05 ± 0.38	164.96 ± 0.34
ColBERT-Clinical ³⁹	79.95 ± 2.06	0.855 ± 0.011	42.29 ± 0.35	157.26 ± 0.33
CMA (Dense RAG) ³⁷	78.71 ± 1.90	0.868 ± 0.011	43.43 ± 0.33	146.46 ± 0.32
MedCPT ³⁸	76.36 ± 1.86	0.875 ± 0.010	42.36 ± 0.32	142.21 ± 0.31
<i>Knowledge Graph Baselines:</i>				
GraphCare ⁴⁰	75.62 ± 1.84	0.871 ± 0.011	41.44 ± 0.33	149.46 ± 0.32
<i>Diagnostic Trajectory Frameworks:</i>				
GDT	68.96 ± 1.57	0.915 ± 0.009	39.47 ± 0.30	128.24 ± 0.28
SDT (Ours)	62.97 ± 1.40	0.939 ± 0.008	34.98 ± 0.28	112.87 ± 0.25

Table 3: **Subgroup stratification across all 9 conditions.** Diagnostic accuracy and mean TTCI (seconds) stratified by Case Complexity (High vs. Low) and Clinician Experience (< 5 years vs. ≥ 5 years).

Condition	Diagnostic Accuracy				Mean TTCI (seconds)			
	Complexity		Experience		Complexity		Experience	
	High	Low	< 5 yr	≥ 5 yr	High	Low	< 5 yr	≥ 5 yr
Control	0.823	0.919	0.870	0.865	121.73	50.92	87.30	90.04
BM25	0.795	0.891	0.845	0.834	124.89	55.24	90.39	94.52
LLM-RAG	0.716	0.866	0.771	0.804	123.98	49.79	89.71	88.53
ColBERT	0.802	0.915	0.865	0.843	111.83	43.86	77.30	83.26
CMA	0.823	0.919	0.870	0.865	108.63	44.84	77.19	80.61
MedCPT	0.825	0.932	0.872	0.879	106.65	42.05	75.34	77.62
GraphCare	0.825	0.923	0.870	0.872	105.00	42.35	74.26	77.30
GDT	0.878	0.957	0.912	0.919	96.22	38.09	68.79	69.16
SDT (Ours)	0.913	0.968	0.935	0.944	87.12	35.62	63.23	62.63

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Table 4: **NASA-TLX multi-dimensional workload subscale means across all 9 conditions.** Scores are on a 0–100 scale. Lower values indicate lower workload on all subscales except Performance (where higher indicates greater perceived success).

Condition	Mental ↓	Physical ↓	Temporal ↓	Perform. ↑	Effort ↓	Frustr. ↓
Control	66.69	18.10	63.70	56.42	64.81	39.89
BM25	67.61	18.53	64.59	55.57	65.71	40.65
LLM-RAG	52.55	15.44	47.82	63.70	50.53	28.22
ColBERT	51.06	14.60	46.64	65.15	49.14	27.15
CMA	52.76	15.13	48.65	64.62	51.02	28.44
MedCPT	51.02	14.48	46.54	65.53	49.18	27.41
GraphCare	49.35	14.25	45.23	66.46	47.36	25.98
GDT	46.17	13.07	41.49	68.45	44.34	23.28
SDT (Ours)	38.34	11.59	33.12	72.76	36.56	17.47

Table 5: **Emergency Department operational impact projected via Erlang C queueing model.** Parameters: $c = 8$ attending physicians, arrival rate $\lambda = 6.0 \text{ hr}^{-1}$, baseline service rate $\mu_0 = 0.80 \text{ hr}^{-1}$. Δt : mean per-patient search time saved (seconds); μ' : adjusted service rate (hr^{-1}); ρ' : traffic intensity; P_W : probability of queuing; W_q : expected queue waiting time in minutes.

Condition	Δt (s)	μ' (hr^{-1})	ρ'	P_W	W_q (min)
Control	0.0	0.8000	0.9375	0.8073	121.10
BM25	0.0	0.8000	0.9375	0.8073	121.10
LLM-RAG	0.0	0.8000	0.9375	0.8073	121.10
ColBERT	8.6	0.8015	0.9357	0.8019	116.74
CMA	9.8	0.8017	0.9354	0.8011	116.14
MedCPT	12.2	0.8022	0.9349	0.7997	114.97
GraphCare	12.9	0.8023	0.9348	0.7993	114.64
GDT	19.6	0.8035	0.9334	0.7952	111.53
SDT	25.6	0.8046	0.9322	0.7915	108.79

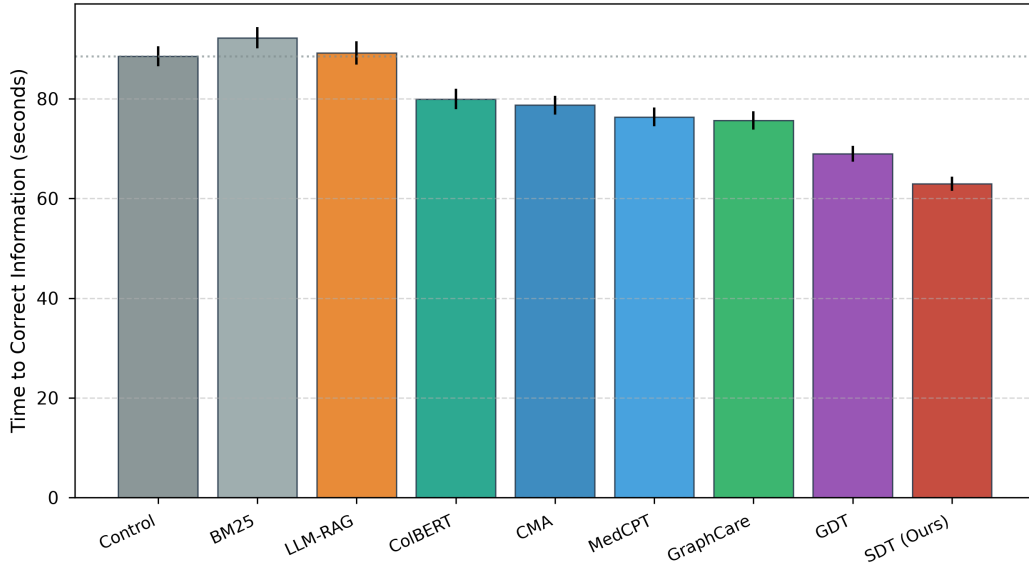


Figure 1: **Mean Time-to-Correct-Information (TTCI) across 9 experimental conditions.** Grouped bar chart showing mean TTCI in seconds $\pm$ standard error across $N = 1,000$ vignettes. SDT achieves 62.97s, reducing search time by 28.9% relative to Control (88.52 s) and 8.7% relative to GDT (68.96 s, paired t -test $p < 10^{-12}$).

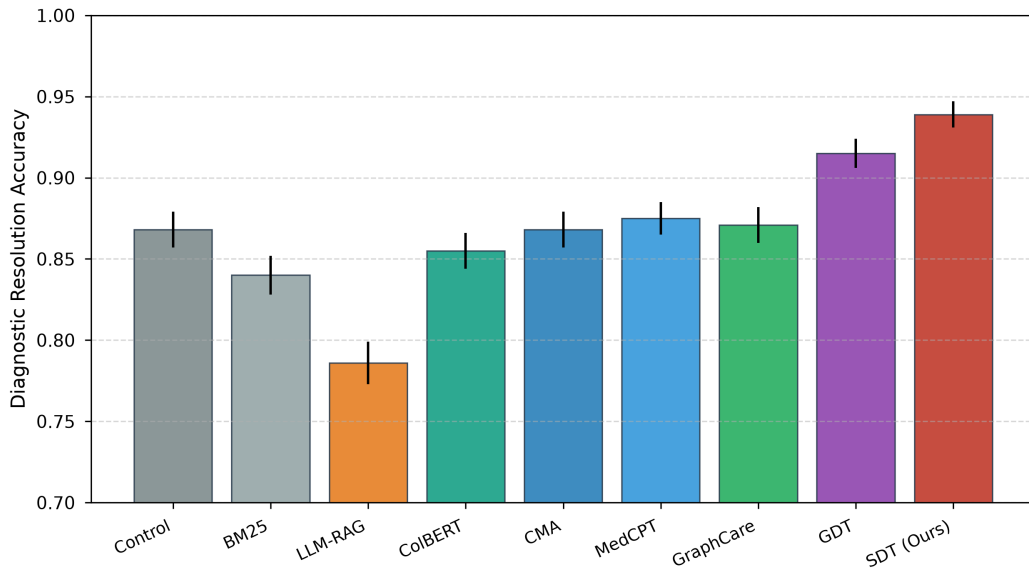


Figure 2: **Diagnostic retrieval accuracy across experimental conditions.** Bar chart showing the proportion of vignettes where confirmatory diagnostic evidence was retrieved within the interaction budget. SDT achieves 93.9% accuracy, compared to 86.8% for Control, 87.5% for MedCPT, and 91.5% for GDT.

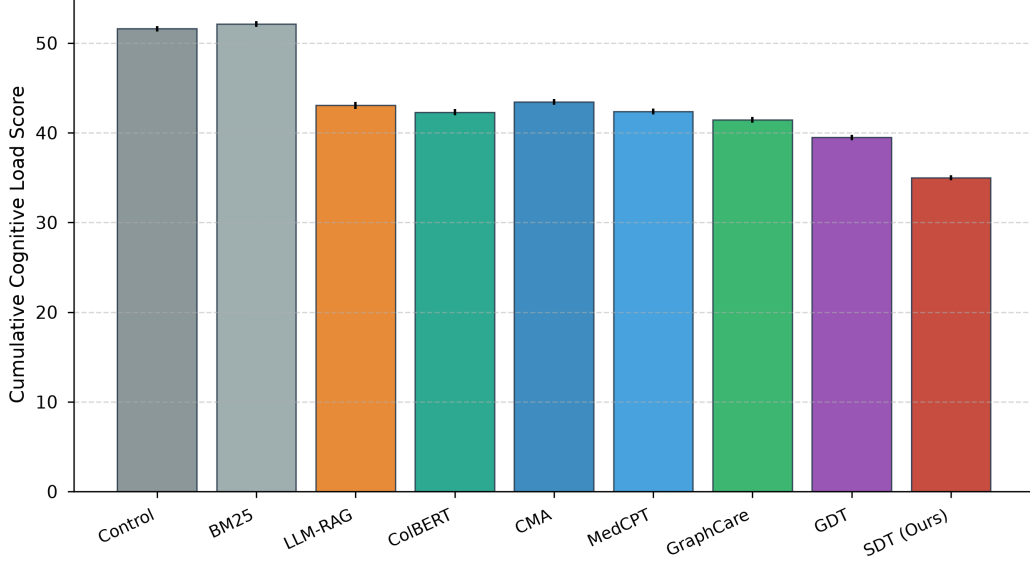


Figure 3: **Modeled cognitive load across experimental conditions.** Modeled cumulative cognitive load (arbitrary units $\pm$ standard error). SDT incurs a mean load of 34.98, a 32.2% reduction compared to Control (51.60) and 11.4% lower than GDT (39.47).

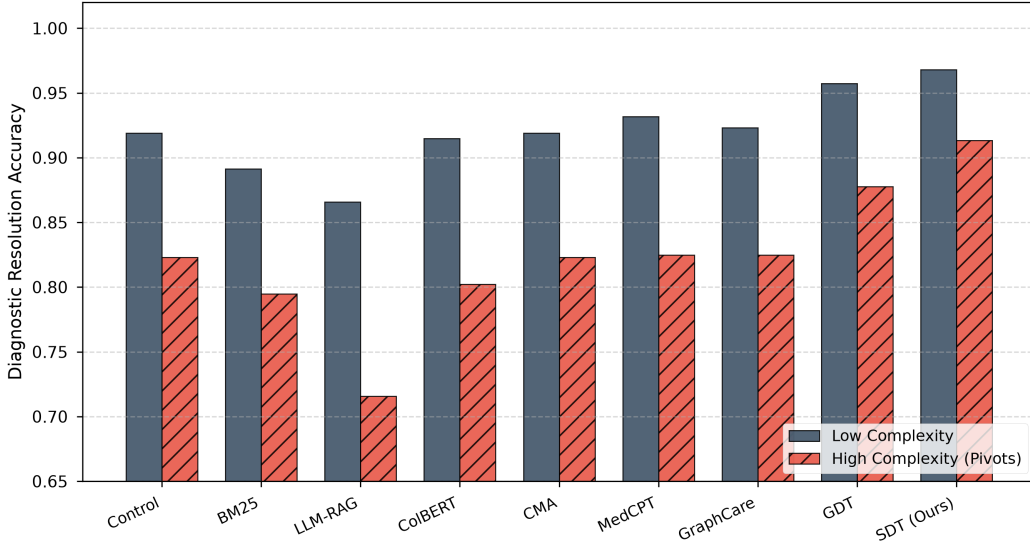


Figure 4: **Diagnostic accuracy stratified by case complexity.** Grouped bar chart depicting retrieval accuracy across Low vs. High case complexity. On high-complexity vignettes, SDT preserves 91.3% accuracy (+9.0 pp over Control and +3.5 pp over GDT), whereas LLM-RAG degrades to 71.6%.

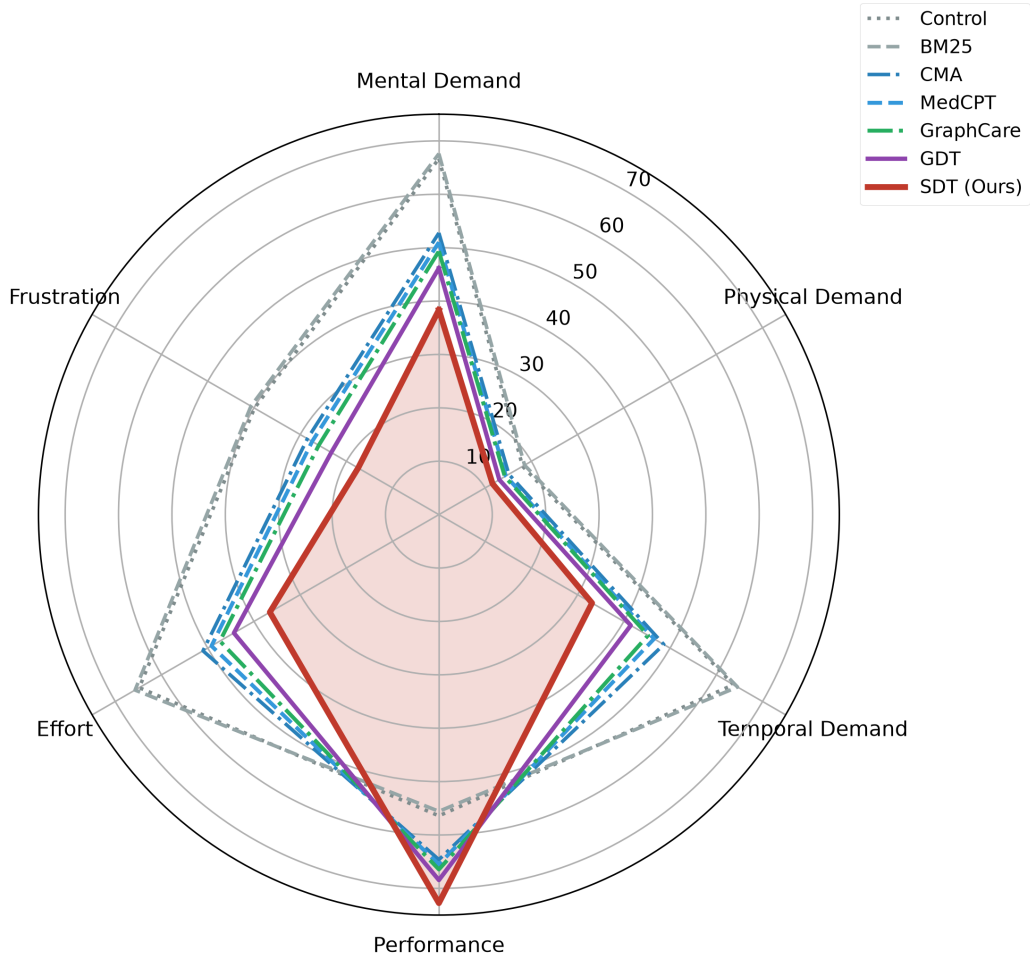


Figure 5: **Multi-dimensional NASA-TLX subjective workload profile.** Radar chart illustrating mean scores across all six NASA-TLX dimensions. SDT exhibits the smallest enclosed workload area, with marked reductions in Mental Demand (38.34 vs 66.69) and Frustration (17.47 vs 39.89), and enhanced Performance (72.76 vs 56.42).

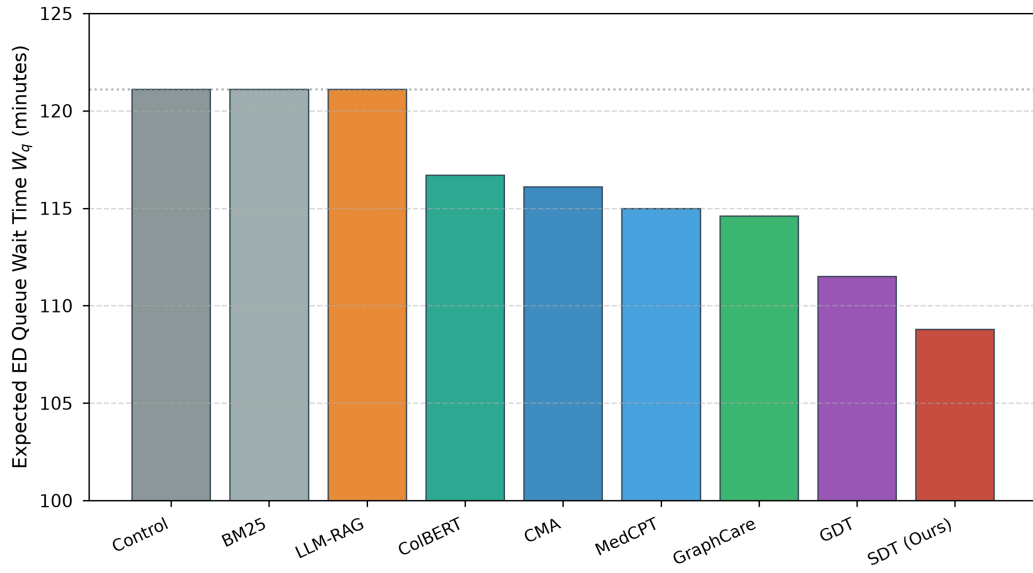


Figure 6: **Emergency Department operational queue waiting time (W_q)**. Projected queue waiting time in minutes under an Erlang C queueing model ($c = 8$ physicians, traffic intensity $\rho_0 = 0.9375$). Saving 25.6 seconds per encounter reduces average patient queue waiting time from 121.10 min to 108.79 min, recovering over 10,200 hours of patient waiting time annually.